

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA0601 COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 20%

QUESTIONS: 15 Total Marks:25 Annexure A

Case Study

Code of Conduct:

I N.HarshaVardhan(192324189) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:N.HarshaVardhan

Annexure A

Short Answer Test

1, A hospital aims to develop an intelligent system to diagnose diseases based on patient symptoms and medical history. Using concepts such as Bayes' Theorem, Naïve Bayes classifiers, and Bayesian Belief Networks, design a probabilistic model for diagnosis. Analyze how uncertainty is handled and how conditional dependencies between symptoms influence predictions. Discuss the role of probability learning and evaluate the effectiveness of your approach with respect to real-world constraints.

Answer:

Probabilistic Model for Intelligent Disease Diagnosis

A hospital can build an intelligent disease diagnosis system using Bayes' Theorem, a Naïve Bayes Classifier, and a Bayesian Belief Network (BBN). These probabilistic techniques estimate the likelihood of diseases based on patient symptoms, medical history, laboratory results, and other clinical information, while effectively handling uncertainty.

1. Model Design

Step 1: Collect Patient Data

The system gathers:

- Symptoms (fever, cough, headache, chest pain)
- Medical history (diabetes, hypertension, allergies)
- Laboratory test results
- Patient demographics (age, gender)
- Lifestyle factors (smoking, alcohol use)

These variables become the evidence used for diagnosis.

Step 2: Disease Prediction Using Bayes' Theorem

Bayes' Theorem calculates the probability of a disease after observing

symptoms. $P(D|S) = \frac{P(S|D)P(D)}{P(S)}$

$$P(D) = 0.85(0.20)$$

$$0.25 = 0.68$$

$$\begin{aligned} &P(D) \\ &P(S) \\ &P(S|D) \\ &P(S|D) \\ &P(S|D) \\ &P(S|D) \\ &P(S|D) \end{aligned}$$

$P(B)=0.25P(B|A)P(A)=0.17P(A|B)\sim 0.68$ Posterior = useful evidence / total evidence
 $P(B)=0.25P(B|A)P(A)=0.17P(A|B)\sim 0.68$ Posterior = useful evidence / total evidence Where:

- $P(\text{Disease} | \text{Symptoms})$ = Posterior probability
- $P(\text{Symptoms} | \text{Disease})$ = Likelihood
- $P(\text{Disease})$ = Prior probability
- $P(\text{Symptoms})$ = Evidence

Example

Suppose:

- Probability of Flu = 0.08
- Probability of Fever given Flu = 0.90
- Probability of Fever = 0.20

Then,

Thus, a patient with fever has a 36% probability of having flu.

Step 3: Naïve Bayes Classifier

The Naïve Bayes classifier assumes that symptoms are conditionally independent given the disease. For example:

Disease = Pneumonia

Symptoms:

- Fever
- Cough
- Chest pain
- Shortness of breath

The classifier computes:

The disease with the highest probability is selected as the predicted diagnosis. Advantages

- Fast computation
- Works well with large medical datasets
- Handles missing values reasonably well
- Suitable for real-time hospital systems

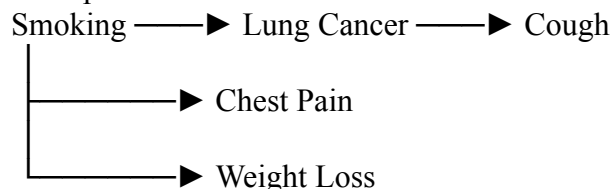
Limitation

The independence assumption is often unrealistic because many symptoms occur together.

Step 4: Bayesian Belief Network (BBN)

A Bayesian Belief Network represents dependencies among diseases and symptoms using a directed graph.

Example:



Family History —> Lung Cancer

Each node represents a variable:

- Risk factors
- Diseases
- Symptoms

Each edge represents a conditional dependency.

Unlike Naïve Bayes, BBNs capture relationships between symptoms.

2. Handling Uncertainty

Medical diagnosis always involves uncertainty because:

- Symptoms overlap among diseases.
- Patients may report symptoms inaccurately.
- Test results may be incomplete.
- Diseases can exist simultaneously.

Probabilistic models manage uncertainty by:

- Assigning probabilities instead of fixed decisions.
- Updating beliefs whenever new evidence becomes available.
- Providing confidence levels for each diagnosis.

Example:

Initially:

- COVID-19 = 25%
- Influenza = 50%
- Pneumonia = 25%

After receiving an X-ray result:

- COVID-19 = 15%
- Influenza = 20%
- Pneumonia = 65%

Thus, the diagnosis becomes more accurate as new information is added.

3. Influence of Conditional Dependencies

Conditional dependency means one symptom may depend on another or both may be caused by the same disease.

In Naïve Bayes

Symptoms are treated as independent.

Example:

- Fever
- Cough

The model assumes:

- Fever does not affect cough.
- Cough does not affect fever.

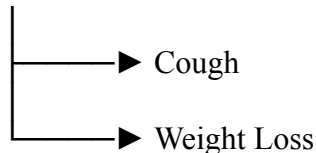
This simplifies calculations but may reduce accuracy.

In Bayesian Belief Networks

Dependencies are explicitly modeled.

Example:

Tuberculosis



Smoking → Cough

Now the model understands:

- Smoking alone can cause cough.
- Tuberculosis can cause both cough and weight loss.
- Observing multiple related symptoms increases confidence appropriately.

Therefore, Bayesian networks usually provide more realistic predictions.

4. Role of Probability Learning

Probability learning estimates the model parameters from historical patient records. Learning involves:

- Disease prevalence (prior probabilities)
- Probability of symptoms given diseases
- Relationships between variables
- Updating probabilities using newly collected hospital data

Methods include:

- Maximum Likelihood Estimation (MLE)
- Bayesian parameter estimation
- Expectation-Maximization (EM) for incomplete data

Benefits:

- Improves diagnostic accuracy over time
- Adapts to new diseases
- Learns local population characteristics

- Reduces manual rule creation

5. Effectiveness Under Real-World Constraints

Advantages

- Handles uncertain and incomplete medical data.
- Produces probabilistic diagnoses rather than simple yes/no answers.
- Supports doctors in decision-making.
- Easy to update with new patient information.
- Can integrate laboratory reports and medical history.
- Naïve Bayes is computationally efficient for large hospitals.

Limitations

- Naïve Bayes assumes symptom independence, which is often unrealistic.
- Bayesian Belief Networks require expert knowledge and large datasets to define accurate structures.
- Rare diseases may lack sufficient training data.
- Incorrect or biased patient records can reduce accuracy.
- Privacy and regulatory requirements (e.g., protection of patient data) must be addressed.
- The system should assist clinicians rather than replace medical judgment.

Conclusion

A probabilistic diagnosis system that combines Bayes' Theorem, Naïve Bayes, and Bayesian Belief Networks provides an effective framework for intelligent medical diagnosis. Bayes' Theorem updates disease probabilities using observed evidence, Naïve Bayes offers fast and scalable classification, and Bayesian Belief Networks capture realistic dependencies among symptoms and risk factors. By learning probabilities from historical patient data and continuously updating predictions as new evidence becomes available, such a system can improve diagnostic accuracy, manage uncertainty, and support clinicians in making informed decisions while acknowledging practical constraints such as data quality, computational complexity, and patient privacy.

2. An organization wants to build a spam detection system for emails using machine learning techniques. Using Naïve Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyze system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.

Answer

Spam Detection System Using Naïve Bayes and Bayes Optimal Classifier 1. Objective

The objective is to design a machine-learning-based email spam detection system that classifies incoming emails into Spam or Not Spam (Ham). The system uses Naïve Bayes for efficient classification and the Bayes Optimal Classifier as a theoretical framework for obtaining the best possible prediction under the assumed probability model.

2. Feature Extraction

Useful email features include:

- Word frequency
- Subject-line words
- Sender information
- Presence of URLs
- Number of attachments
- Special characters

- Words such as "winner", "offer", "free", "prize"
- HTML content
- Email length

The TF-IDF representation or word-frequency representation can be used to convert emails into numerical features.

3. Naïve Bayes Classifier

Naïve Bayes uses Bayes' theorem to calculate the probability that an email belongs to a particular class.

$P(B)=0.25$
 $P(B|A)P(A)=0.17$
 $P(A|B)\sim 0.68$
 Posterior = useful evidence / total evidence
 For an email containing features :

where:

- = Spam or Ham
- = prior probability of the class
- = probability of a feature occurring in that class
- = posterior probability

The class with the highest posterior probability is selected.

Example

Suppose an email contains:

Congratulations! You won a free prize.

Words such as free, won, and prize may have high probabilities in spam emails. Therefore:

$P(\text{Spam} | \text{Email}) > P(\text{Ham} | \text{Email})$

and the email is classified as Spam.

Advantages of Naïve Bayes

- Fast training and prediction
- Works well with high-dimensional text data
- Requires relatively small training datasets
- Easy to implement
- Low computational cost

Limitation

Naïve Bayes assumes that features are conditionally independent. In real emails, words are often related to each other, so this assumption may not always hold.

4. Maximum Likelihood Estimation vs MDL

Feature selection is important because emails can contain thousands of possible words and attributes.

Maximum Likelihood Estimation (MLE)

MLE selects parameter values that maximize the likelihood of observing the training data. For spam detection, MLE can estimate how frequently a particular word occurs in spam and ham emails.

Advantages

- Simple and efficient
- Easy to calculate
- Works well with large datasets
- Useful for estimating word probabilities

Disadvantages

- Can overfit training data
- Does not directly penalize model complexity
- Rare features can produce unreliable estimates

Minimum Description Length (MDL)

The MDL principle selects a model that provides the shortest total description of:

1. The model itself
2. The data encoded using that model

Conceptually:

The best model minimizes this total description length.

For spam detection, MDL can remove unnecessary features and retain features that provide useful information for classification.

MLE vs MDL

Aspect MLE MDL

	Main objective	Maximize likelihood	Minimize total description length
Complexity			
penalty	No direct penalty	Yes	
Overfitting	More susceptible	Reduced	
Aspect	MLE	MDL	
			Balances usefulness and complexity
Feature selection	Based mainly on predictive likelihood		
Model size	Can become large	Usually more compact	Computation Lower
Generally higher			

5. Real-World Considerations

A practical spam detection system must handle:

- New types of spam
- Changing spam vocabulary
- Imbalanced spam/ham datasets
- Obfuscated words such as fr33, w1nner
- HTML and malicious links
- Multilingual emails
- False positives
- Privacy concerns
- Continuous changes in attacker behavior

The model should therefore be periodically retrained using newly labeled emails.

Conclusion

A spam detection system based on Naïve Bayes provides a fast, scalable, and effective solution for classifying emails. The Bayes Optimal Classifier provides a theoretical benchmark by considering uncertainty across possible hypotheses. Maximum Likelihood Estimation is efficient for learning feature probabilities, whereas MDL additionally controls model complexity and helps avoid unnecessary features. Larger training datasets generally improve probability estimates and classification accuracy, while an excessively large

hypothesis space can cause overfitting. Therefore, an effective real-world system should balance accuracy, model complexity, training-data size, computational cost, and false-positive risk.

3. A telecommunications company aims to predict customer churn using historical usage data and service feedback. Using probabilistic models such as **Naïve Bayes**, **Bayesian inference**, and hypothesis space search, design a predictive system. Analyze how prior probabilities and feature dependencies influence churn prediction. Evaluate how uncertainty in customer behavior is handled and discuss the effectiveness of the model in improving customer retention strategies.

Answer

Probabilistic Customer Churn Prediction

1 Problem Statement

A telecommunications company wants to predict which customers are likely to discontinue their services. The company has historical customer information such as:

- Customer tenure
- Monthly charges
- Call and internet usage
- Number of complaints
- Service feedback
- Contract type
- Payment history
- Previous churn status

The company needs a probabilistic system that can identify high-risk customers and support effective customer retention strategies.

2 Objectives

- Predict whether a customer is likely to churn.
- Calculate the probability of customer churn.
- Use historical data as prior knowledge.
- Handle uncertainty in customer behavior.
- Analyze dependencies between customer features.
- Help the company develop targeted retention strategies.

3 Proposed Probabilistic Model

The system uses three major approaches:

1. Naïve Bayes

- Calculates the probability of churn based on customer features.
- Assumes features are conditionally independent given the churn class.
- Provides fast predictions for large customer datasets.

2. Bayesian Inference

- Starts with a prior probability of churn.
- Updates the probability when new customer information becomes available.
- Helps the system adapt to changing customer behavior.

3. Hypothesis Space Search

- Generates different possible churn hypotheses.
- Examples:
 - High complaints → High churn probability.
 - Low usage + poor feedback → High churn probability.
 - Short tenure + expensive plan → High churn probability.
- Selects the hypothesis that performs best on historical data.

4 Role of Prior Probability

Prior probability represents the historical likelihood of churn.

For example, if 20% of existing customers have previously left the company: When new evidence is received, Bayesian inference updates this probability: Thus, historical churn information provides the initial knowledge, while current customer information modifies the prediction.

5 Feature Dependencies

Some customer features are related to each other.

For example:

Poor Network Quality → Customer Dissatisfaction → Complaints → Churn Naïve

Bayes treats features as conditionally independent, which may not always represent real world relationships accurately.

A Bayesian network can model these dependencies more effectively and improve prediction when relationships between features are important.

6 Handling Uncertainty

The model does not simply provide a Yes/No prediction. It can provide a probability.

For example:

- Churn probability = 15% → Low risk
- Churn probability = 50% → Medium risk
- Churn probability = 85% → High risk

This allows the company to prioritize customers according to their risk level.

7 Evaluation

The system can be evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- Confusion matrix

Recall is especially important because identifying customers who are likely to churn allows the company to take preventive action.

8 Retention Strategies

Based on the predicted churn probability, the company can:

- Provide personalized discounts.
- Offer improved service plans.
- Contact dissatisfied customers.
- Provide loyalty rewards.
- Improve customer support.
- Offer customized packages.

9 Challenges

- Customer behavior changes over time.
- Features may not be completely independent.
- Historical data may contain missing or incorrect values.
- Imbalanced churn data can affect prediction.
- Customer preferences can be difficult to predict accurately.

Conclusion

The probabilistic churn prediction system provides an effective method for identifying

customers who are likely to leave. Naïve Bayes provides fast prediction, while Bayesian inference updates predictions using new information. Hypothesis space search helps identify useful churn patterns. By handling uncertainty through probability-based predictions, the telecommunications company can identify high-risk customers early and apply targeted retention strategies.

4. A smart healthcare system is designed to monitor patients in real-time using wearable sensors. Using **Bayesian Belief Networks** and probability learning, develop a model to detect anomalies in vital signs. Analyze how conditional dependencies between physiological parameters are modeled and how uncertainty due to sensor noise is handled. Evaluate system performance and discuss challenges in real-time deployment.

Answer:

Real-Time Healthcare Anomaly Detection Using Bayesian Belief Networks

1 Problem Statement

A smart healthcare system uses wearable sensors to continuously monitor patients. The sensors collect information such as:

- Heart rate
- Blood pressure
- Body temperature
- SpO₂
- Respiratory rate
- ECG
- Physical activity

The challenge is to identify abnormal health conditions from noisy and continuously changing sensor data.

2 Objectives

- Monitor patients continuously.
- Detect abnormal vital signs.
- Model relationships between physiological parameters.
- Handle sensor noise and missing data.
- Provide real-time alerts.
- Reduce unnecessary false alarms.

3 Proposed Bayesian Belief Network

A **Bayesian Belief Network (BBN)** is used to represent dependencies between physiological variables.

A simplified structure is:

Physical Activity → Heart Rate → Blood Pressure → Health Condition

Another relationship can be:

Respiratory Condition → Respiratory Rate → SpO₂ → Health Condition

Each node contains probability values learned from historical patient data. 4

Probability Learning

The system learns conditional probabilities from historical healthcare data.

For example:

may be high because exercise naturally increases heart rate.

However:

may indicate a possible abnormal condition.

Therefore, the system considers the patient's overall condition instead of evaluating one sensor reading independently.

5 Conditional Dependencies

Physiological parameters are strongly related.

For example:

- Physical activity affects heart rate.
- Heart rate can be related to blood pressure.
- Respiratory problems can reduce SpO₂.
- Fever can increase heart rate and body temperature.

The Bayesian network represents these dependencies and calculates the probability of an abnormal condition based on multiple observations.

6 Handling Sensor Noise

Wearable sensors may produce inaccurate readings because of:

- Patient movement
- Poor sensor contact
- Battery problems
- Communication errors
- Environmental interference

Bayesian reasoning handles this uncertainty by assigning probabilities rather than assuming every measurement is completely accurate.

For example:

If this probability exceeds a predefined threshold, the system can generate an alert.

7 Real-Time Working

The system operates as follows:

1. Wearable sensors collect patient data.
2. Sensor data is transmitted to the monitoring system.
3. Noise and missing values are processed.
4. The Bayesian network receives the observations.
5. Conditional probabilities are calculated.
6. The probability of an abnormal condition is determined.
7. An alert is generated when the probability exceeds the threshold.
8. New sensor readings continuously update the prediction.

8 Performance Evaluation

The system can be evaluated using:

- Accuracy
- Sensitivity
- Specificity
- Precision
- F1-score
- False alarm rate
- Detection time

High sensitivity is important because the system should detect dangerous conditions as early as possible.

9 Challenges

- Continuous sensor data produces a large computational workload.
- Sensor readings can be noisy.
- Missing data can affect predictions.
- Patients have different normal vital-sign ranges.
- Excessive false alarms can cause alert fatigue.
- Patient data must be protected.
- Real-time processing requires low latency.

10 Conclusion

A Bayesian Belief Network is suitable for real-time healthcare monitoring because it represents conditional dependencies between physiological parameters and handles uncertainty in sensor observations. Probability learning allows the system to improve its predictions using historical data. The system can provide early warnings and support healthcare professionals, although accuracy, privacy, real-time processing, and false alarms must be carefully managed.

5. A logistics company wants to optimize delivery routes under uncertain traffic and

weather conditions. Using **Bayesian inference** and probabilistic reasoning, design a system that predicts optimal routes dynamically. Analyze how prior knowledge and real-time data updates influence decision-making. Evaluate computational complexity and discuss how uncertainty impacts route optimization efficiency.

Answer

Dynamic Delivery Route Optimization Using Bayesian Inference

1 Problem Statement

logistics company needs to deliver packages efficiently, but travel time is uncertain because of:

- Traffic congestion
- Accidents
- Weather conditions
- Road closures
- Road conditions
- Vehicle delays

The company requires a system that can dynamically select the best route as conditions change. **2 Objectives**

- Predict travel time under uncertain conditions.
- Combine historical and real-time information.
- Select the optimal delivery route.
- Dynamically update routes.
- Reduce delivery time and transportation costs.
- Improve on-time delivery performance.

3 Proposed Probabilistic Model

The system uses:

1. Bayesian inference

- Uses historical traffic and weather information as prior knowledge.
- Updates probabilities using real-time observations.

2. Probabilistic reasoning

- Represents different possible traffic and weather conditions.
- Calculates the expected cost of each route.

3. Route optimization

- Selects the route with the lowest expected travel cost.

4 Role of Prior Knowledge

Historical data provides prior probabilities.

For example, if a particular road frequently experiences congestion between 5 PM and 7 PM: will be relatively high.

This prior knowledge helps the system make an initial route decision even before receiving current traffic information.

5 Real-Time Bayesian Updating

When new information becomes available, the system updates the previous probability.

For example:

- Initially, Route A = 30 minutes.
- Route B = 35 minutes.
- Route A is selected.

Later, an accident is reported on Route A.

The system updates the probability of delay on Route A. If the expected travel time becomes 50 minutes, Route B may become the optimal choice.

Thus:

Historical Data + Real-Time Data → Bayesian Update → New Route Decision

6 Probabilistic Route Selection

The expected cost of a route can be calculated as:

Possible states include:

- Normal traffic
- Heavy traffic
- Rain
- Accident
- Road closure

The route with the lowest expected cost is selected.

7 Handling Uncertainty

Instead of assuming that a route will always take a fixed amount of time, the system considers different possibilities.

For example:

Travel Time Probability

30 minutes 20%

40 minutes 50%

55 minutes 30%

This allows the system to make decisions even when exact travel time is unknown. **8 System Working**

The system follows these steps:

1. Collect historical traffic and weather data.
2. Collect current GPS and traffic information.
3. Estimate prior probabilities.
4. Update probabilities using real-time evidence.
5. Calculate expected travel cost for each route.
6. Select the optimal route.
7. Continuously monitor traffic and weather.
8. Reroute the vehicle when necessary.

9 Computational Complexity

For a road network with locations and roads, shortest-path algorithms such as Dijkstra's algorithm can have approximately:

complexity when implemented with a priority queue.

Bayesian updates add additional computational requirements. For large networks with frequent real time updates, the system may require:

- Approximate inference
- Caching
- Parallel processing
- Hierarchical routing
- Efficient shortest-path algorithms

10 Performance Evaluation

The system can be evaluated using:

- Average delivery time
- On-time delivery percentage
- Fuel consumption
- Distance travelled
- Number of reroutes
- Route prediction accuracy
- Computational time
- Transportation cost

11 Challenges

- Traffic conditions change rapidly.
- Weather information may be uncertain.
- Real-time data may contain errors.
- Frequent rerouting can increase computational cost.

- Large road networks require efficient algorithms.
- Poor network connectivity can delay real-time updates.

12 Conclusion

Bayesian inference provides an effective approach for dynamic delivery route optimization because it combines historical knowledge with real-time information. The system can update route probabilities whenever traffic or weather conditions change and select the route with the lowest expected cost. Although uncertainty and computational complexity create challenges, probabilistic reasoning can improve delivery efficiency, reduce delays and fuel consumption, and increase customer satisfaction.

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Clarity & Presentation	15	Clear, concise, and well-organized answer	Under e wit issues

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (Marks)
Conceptual Understanding	30	Demonstrates clear complete understanding the concept
Accuracy of Answer	20	Completely correct answers with no errors
Application of Knowledge	20	Correctly apply concepts to solve the question/problem
Completeness of Response	15	Covers all required parts the question